Differential Gene Expression Analysis between BRCA1 mutant ER negative vs normal breast samples: Bulk RNA-seq analysis using a publicly available dataset

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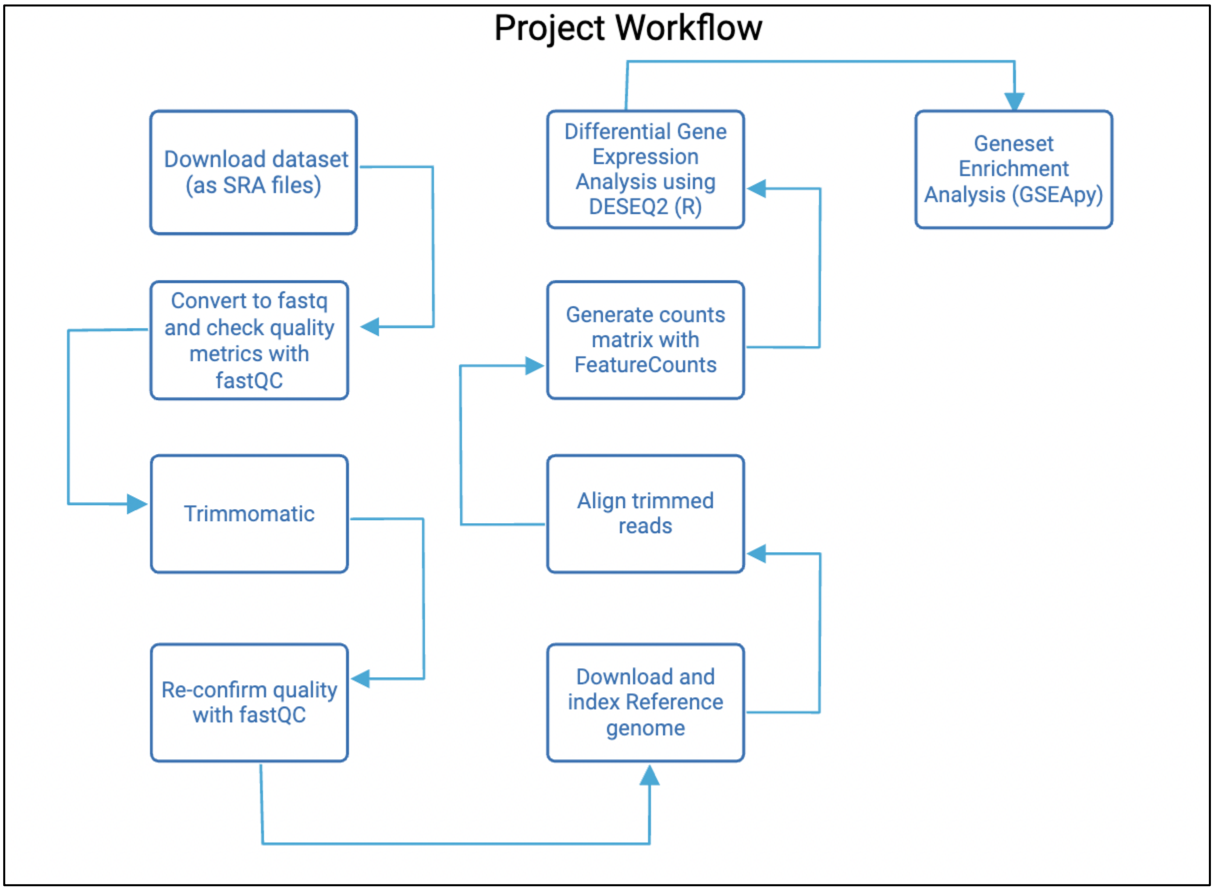
(Link to Github documentation)

OBJECTIVE:

The objective of this study was to:

a) Understand the differentially expressed genes between nine patient tumor samples harbouring *BRCA1* mutation and negative for estrogen receptor (ER) status, versus 6 normal breast tissue samples.
b) To identify biological pathways that are upregulated/downregulated in *BRCA1* mutant ER negative patients.
c) To learn the bioinformatics tools and techniques used in this analysis, from downloading the dataset using the command line, DESeq2 analysis using R, and GSEApy in Python.

WORKFLOW (ABSTRACT):



INTRODUCTION:

Breast cancer is one of the leading causes of cancer, in women accounting for 11.6% of all cancers and is the fourth highest cause of cancer-associated mortality worldwide [1]. Of these, hereditary breast cancers (HBC), which occurs due to inherited genetic mutations in high-penetrance genes, contribute to nearly 10% of all breast cancer cases, and is associated with early onset of disease. Some of these genes include known *BRCA1, BRCA2, TP53, PTEN*, etc. that are known tumor suppressor genes and are involved in processes such as DNA repair and cell cycle [2]. Carriers of *BRCA1* and *BRCA2* mutations have a significantly higher lifetime risk of developing breast cancer (BC) and ovarian cancers (OC). Mutations in *BRCA1* are also associated with triple negative breast cancer, a particularly aggressive breast cancer subtype characterized by the absence of estrogen (ER), Progesterone (PR) and HER2 receptors [3]. Therefore, identifying genes that are expressed differentially in patients with germline mutations in *BRCA1,* could point to important molecular targets that can be used to inform therapeutic strategies.

The dataset analyzed in this study was obtained from Shah et. al. (2022) [4]. In the original study, 67 tumor samples were collected from 13 *BRCA1*/2 germline mutation carriers with BC and 14 *BRCA1/2* germline mutation carriers with OC. Of the 13 BC cases, 13 primary tumors and 18 recurrent tumors were collected and of the 14 OC cases, 22 recurrent tumor samples were collected in addition to the 14 primary tumors. The focus of the original study was multi-omic analysis of primary and recurring tumors to identify the molecular drivers of breast cancer recurrence. One of the main outcomes of the original study was the observation that PARP1 was amplified and overexpressed in *BRCA1/2* mutant primary and recurrent tumors. For the present study, only a subset of all the samples, that had both BRCA1 mutation, and additionally were ER-ve, and normal breast samples were selected for this study. The results of the study corroborate with the existing knowledge of differentially expressed genes in *BRCA1* mutant breast cancers and underscore the power of employing such methods in informing therapeutic strategies for patients.

METHODS:

Study Description and Downloading the dataset

In this study, the primary tumors of nine *BRCA1* germline mutation carriers that also had estrogen receptor negative (ER-ve) status (characterized as predominantly TNBC) were selected. Normal breast tissue from six patients, who are carriers of *BRCA1/2* mutations with no history of cancer or cancer treatment were selected as the control group (see associated excel file 'SAMPLE.xlsx' in Data for sample details).

The RNA-seq files for the selected samples were obtained from the NCBI SRA database (Accession number PRJNA751555) [4]. The relevant samples (as indicated in 'SAMPLE.xlsx' in Data) were downloaded using prefetch version 3.2.0. All the relevant details such as sample information, coding scripts, and results are documented on Github.

Trimming and Quality check

The SRA files were converted to FastQ format using fasterq-dump version 3.2.0 and a preliminary quality check was performed using fastQC (v0.11.9). The reads were trimmed initially with the following parameters: seed mismatch allowed was set at 2, palindrome clip threshold was set at 30 and simple clip threshold was set at 10. Leading and trailing was set at 3, with sliding window 4:15 and minlen 36. These parameters were later changed in the following way: palindromic clip threshold 20, leading and trailing 2, sliding window 4:12 and minlen 30. The parameters were made slightly less aggressive because there was a sharp drop in the number of reads post trimming. A comparison in reads from both parameters are given in the excel file 'QC_metrics.xlsx'.

Reference genome indexing

The GrCh37.p13 genome assembly and the corresponding GENCODE v19 annotation file (specifically, GRCh37.p13.genome.fa and gencode.v19.annotation.gtf) were used. This was indexed using STAR (version 2.7.10b) with overhang set at 149.

Alignment and counts matrix

The trimmed fastq files were aligned to the indexed reference genome using STAR aligner (v2.7.10b). The output file of the aligned reads was BAM, and the aligned reads were sorted by coordinate. Read counts were obtained at the gene level using featureCounts (v2.0.3) using the GENCODE v19 gene annotation gtf file. Coordinate sorted paired end BAM files that were generated after STAR alignment of the reads was used as input.

Differential Gene Expression Analysis

The counts matrix generated with featureCounts was used in differential gene expression analysis using DESeq2 (version 1.34.0) in R (version 4.1.2). lfcshrink() function was used for downstream differential gene expression analysis, and parallely, unshrunken DESeq2 results were generated for GSEA analysis. The metadata was set to identify the mutant and control samples. Gene identifiers (Ensembl gene IDs) were mapped to gene symbols using the `gene_name` field from the GENCODE v19 annotation GTF file (`gencode.v19.annotation.gtf`). The GTF was parsed to extract mappings between `gene_id` and `gene_name` attributes.

Gene set enrichment analysis

Gseapy (v1.1.8) was used to perform gene set enrichment analysis. Unshrunken results from DESeq2, which retained the Wald statistic (stat) column was used to rank the genes prior to GSEA.

RESULTS AND DISCUSSION

Sample Quality check (QC)

After the SRA files were converted to fastq format, fastqc was run, to estimate the read depth of the RNA-seq samples, presence of adapters and other quality control parameters. The total number of sequences in each of the paired end reads and unpaired reads matched the number of spots sequenced (as indicated in the sample metadata). Phred Quality score was >30. For all the samples, the per base sequence content and adapter content metrics were flagged, as the presence of Illumina universal adapter was detected towards the 3' end. The samples therefore required trimming.

The per base sequence content shows the average percentage base composition, at each position of the read across all samples. Since humans have 40-50% GC content, approximately equal distribution of base composition can be expected. Any deviation from this norm likely indicates biases due to adapter contamination, PCR amplification bias, etc. In the samples of the current study, the per base sequence content shows skewed base distribution at the 5' end, and could be due to base biases at the start of sequencing.

Trimming and post-trimming (QC)

Trimming was performed using Trimmomatic (v 0.39). As indicated in the methods, section, the original trimming patterns caused a sharp drop in the number of reads retrieved post trimming, as evidenced by running FastQC on the trimmed samples. Adjusting the trimming parameters to be slightly less aggressive, resulted in a minor increase in the number of reads for some samples, and no change in others (see 'QC_metrics.xlsx'). After trimming, the number of read pairs retrieved were in the range of 0.4 – 8.3 million. This is generally considered very low to moderate coverage, with 10-30 million read pairs per sample being standard for optimal differential gene expression analysis.

Post trimming, the quality metrics that were flagged were 'Per base sequence content' and sequence duplication level. The per base sequence content continues to remain flagged post trimming and may be due to the nature of the sequencing protocol. In the current study, Illumina Truseq3 platform was used, which uses random hexamers for generating the cDNA library. Random hexamers have been shown to introduce bias at the 5' end[5], which could be reflected in this flagged parameter. The high sequence duplication level suggests that the library may not be very complex, or that the sample has been over sequenced, both of which may be likely because the RNA sample was derived from Formalin Fixed Paraffin embedded (FFPE) samples. While it is possible that this high level of duplication reflects underlying biological differential expression, it is unlikely, considering the low depth of library coverage and complexity, and more likely appears to be due to technical bias.

Reference genome indexing

The reference genome (GRCh37.p13.genome.fa) and annotation file (gencode.v19.annotation.gtf) were downloaded from https://www.gencodegenes.org/human/release_19.html, and indexed using STAR aligner in the 'GenomeGenerate' mode. The annotation file is compatible with GRCh37.p13, as indicated on the GENCODE webpage, and it was further confirmed that the chromosome naming conventions matched in the fasta and gtf files. As these were 150bp reads, the sjdb overhang parameter was set to 149.

Alignment and counts matrix

After the reference genome had been indexed, the nine tumor and six normal samples were aligned using STAR. The uniquely mapped genes for each sample were between 65-85% for all the samples, with normal samples closer to 85%. The paired_1 and paired_2 trimmed files were used in the alignment, and the output was a bam file sorted by coordinates, which was used as the input for performing gene level quantification using featureCounts. The GENCODE v19 annotation file (gtf) was used to assign genomic features to the aligned output. The -t exon and -g gene_id options were used to read the counts at the exon level and grouped by gene id. -p and –countReadPairs flags were used to ensure proper handling of paired end reads. The featureCounts output was a raw gene count matrix, with many of the reads successfully assigned to annotated features.

Differential Gene Expression Analysis

The raw featureCounts output was cleaned up to prepare for DESeq2 analysis in R. The column names were cleaned to reflect the sample ids, and the relevant columns with the counts for each sample and the gene id column were taken, excluding all the metadata columns. The metadata defining the BRCA1 mutant samples (the first 9 sample columns) and the normal samples (last 6 sample columns) were defined with the normal samples set to be the reference for comparison. The cleaned dataframe and metadata were used to compute the DESeq analysis. The gtf file was parsed to extract gene names using rtracklayer and this was joined with the dataframe to assign gene symbols to the gene ids. Two complementary approaches were used to derive results from DESeq2. Log2 fold change shrinkage (lfcshrink() function) was applied with the apeglm method, particularly to account for genes with low counts or high dispersion. This was used for visualization with heatmaps and volcano plot (Figures 1-3), and to generate the list of most significantly up and downregulated genes.

The results were saved as a csv file and used to filter the most significantly (padj < 0.05) upregulated (log2FC > 1) and downregulated genes (log2FC < -1), the results were saved as a csv file and also represented as a volcano plot and heatmap (Figures 1 – 3). In figure 1, the heat map represents the top 100 most variable genes based on their normalized expression value. This helps us get an overall sense of the data and to visualize whether the normal and mutant samples cluster in distinct patterns. We observe a partial clustering of normal samples vs mutant samples, especially with respect to genes such as TXNIP, BAZ2B, TGFBR3. Figure 2 represents the heatmap generated based on the most significantly differentially expressed genes. The most significantly upregulated genes include *NUF2, CENPF, MTHFD2, STARD7, ASPM.* In a 2020 study by Zhai et. al., the authors analysed potential biomarkers of TNBC using three independent datasets from Gene Expression Omnibus. They found that the top ten hub genes obtained from PPI network analysis included *NUF2, KIF14, KIF15, DEPDC1*, which were also found significantly upregulated in the current study, indicating that these genes may play important regulatory roles in breast cancer biology [6]. *NUF2* codes for a protein that is a component of the NDC80 kinetochore complex and is a crucial player in the segregation of chromosomes during cell division. In fact, several reports exist, that suggest that NUF2 plays a central role in promoting breast cancer [7,8]. *CENPF* is also involved in chromosome segregation during cell divison. Upregulation of such genes indicates improper chromosome segregation, spindle formation and cell cycle progression. *MTHFD2* is a mitochondrial enzyme that is involved in the folate pathway, and is particularly observed to be upregulated in cancers. Similarly, many studies show the upregulation of *STARD7* and ASPM in breast cancers [9,10]. However, the current study did not find upregulation of PARP1, unlike the original study, possibly because of a smaller sample size.

The most significantly downregulated genes in BRCA1 mutant ER negative patients include *PPARG, CRIM1, PPAP2B, DPT, BAZ2B, TBX15*. The low expression of *PPARG* in breast cancers, particularly in ER-ve BCs is well documented [11]. Similarly, CRIM1, DPT, and TBX15 are also known to be downregulated in BCs [12–14]. Interestingly, in a 2023 study that analyzed epigenetic regulators in breast cancer, BAZ2B appeared as one of the key master epigenetic regulators that was downregulated at the protein and mRNA level [15]. Taken together, the differentially expressed genes found in this study, appear to agree with the broader literature on the topic. Fig 3 summarizes the differentially expressed genes between the two groups, represented as a volcano plot.

Gene set enrichment analysis

In DESeq2, in parallel, unshrunken results were extracted for downstream GSEA analysis, which also contains the Wald test (stat) statistic, a reliable parameter for ranking the genes based on differential expression confidence, as this parameter combines log2FC and standard error. The results of DESeq2 were converted to a data frame with a selection of the most relevant columns (gene name and stat). Two gene set collections from MSigDB (Hallmark and Canonical) were provided as gmt files. GSEA was performed using GSEApy with 1000 permutations for statistical rigor. Figure 4 shows the bar plot depicting the top 20 most significantly enriched (10 upregulated and 10 downregulated) as per GSEA with the hallmark gene set. Genes associated with MTORC1 signaling, Myc, G2M Checkpoint, E2F targets and glycolysis are the most significantly enriched as upregulated pathways. The GSEA results suggest that BRCA1 mutant ER-ve tumors may promote tumorigenesis through an upregulation of pathways associated with cell cycle progression, MTORC1 signaling and E2F, a finding that generally agrees with previous analyses in the field [16,17]. However, a downregulation of genes associated with IL6/JAK/STAT3 signaling and inflammatory signaling was observed, potentially reflecting tumor heterogeneity with respect to immune response [18]. These immune evasive phenotypes are also associated with increased glycolysis as observed in this study.

Conclusion:

In this study, the differentially expressed genes between nine patient tumor samples harboring *BRCA1* mutation and ER-ve status versus six normal breast tissue samples was investigated. Although the library depth of coverage was low to moderate, this could be due to loss of information as the original study utilized FFPE samples. Nevertheless, the results suggest an upregulation of genes associated with chromosome segregation, spindle formation, and cell cycle progression, and a downregulation of genes associated with inflammatory response, and factors that can suppress invasion and migration. Overall, the findings of this study recapitulate the consensus in the literature, indicating successful deployment of the bioinformatics tools.

Figures and Figure Legends:

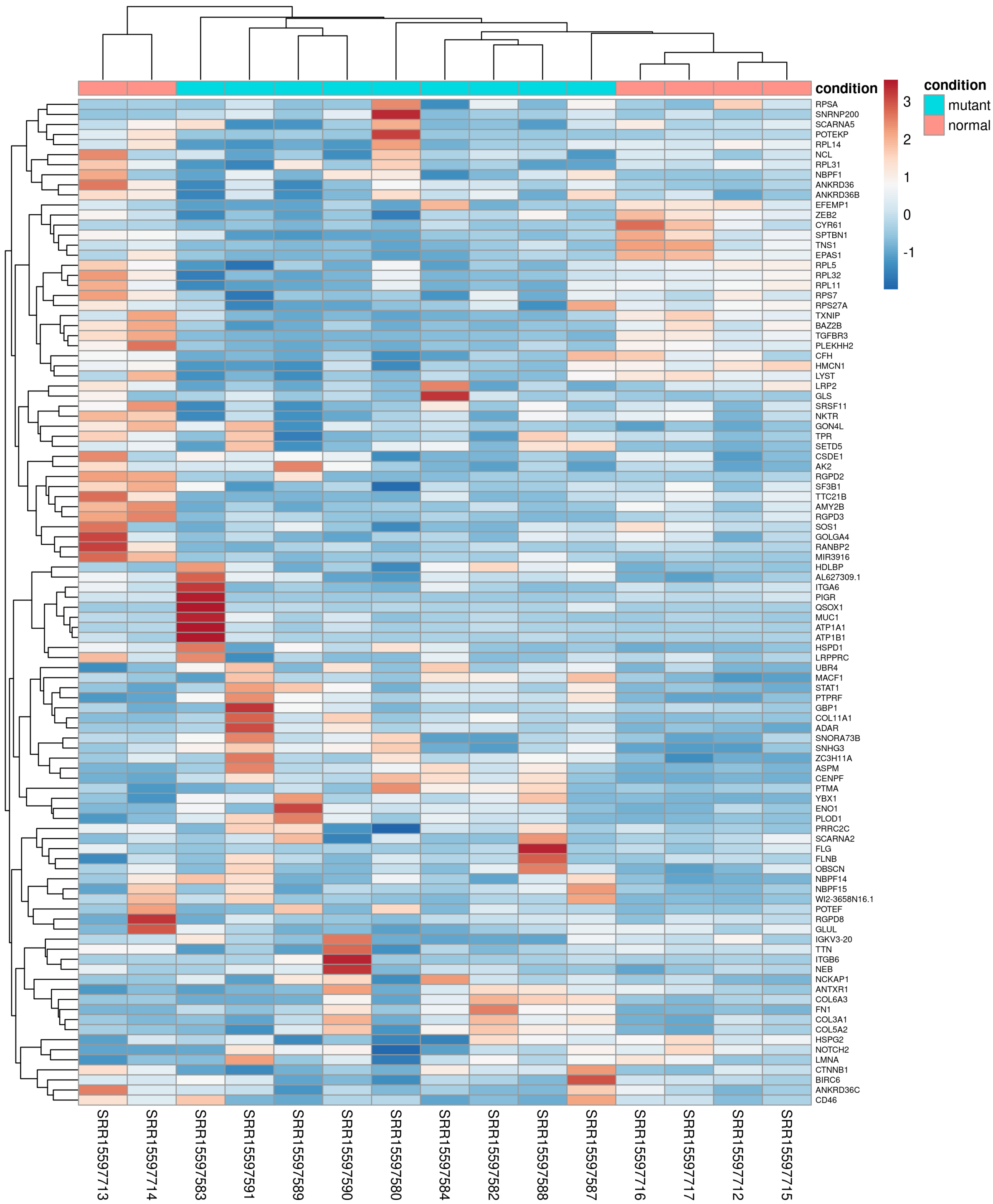


Fig 1: Heatmap based on Top 100 most variable genes between BRCA1 mutant ER -ve tumors vs normal samples

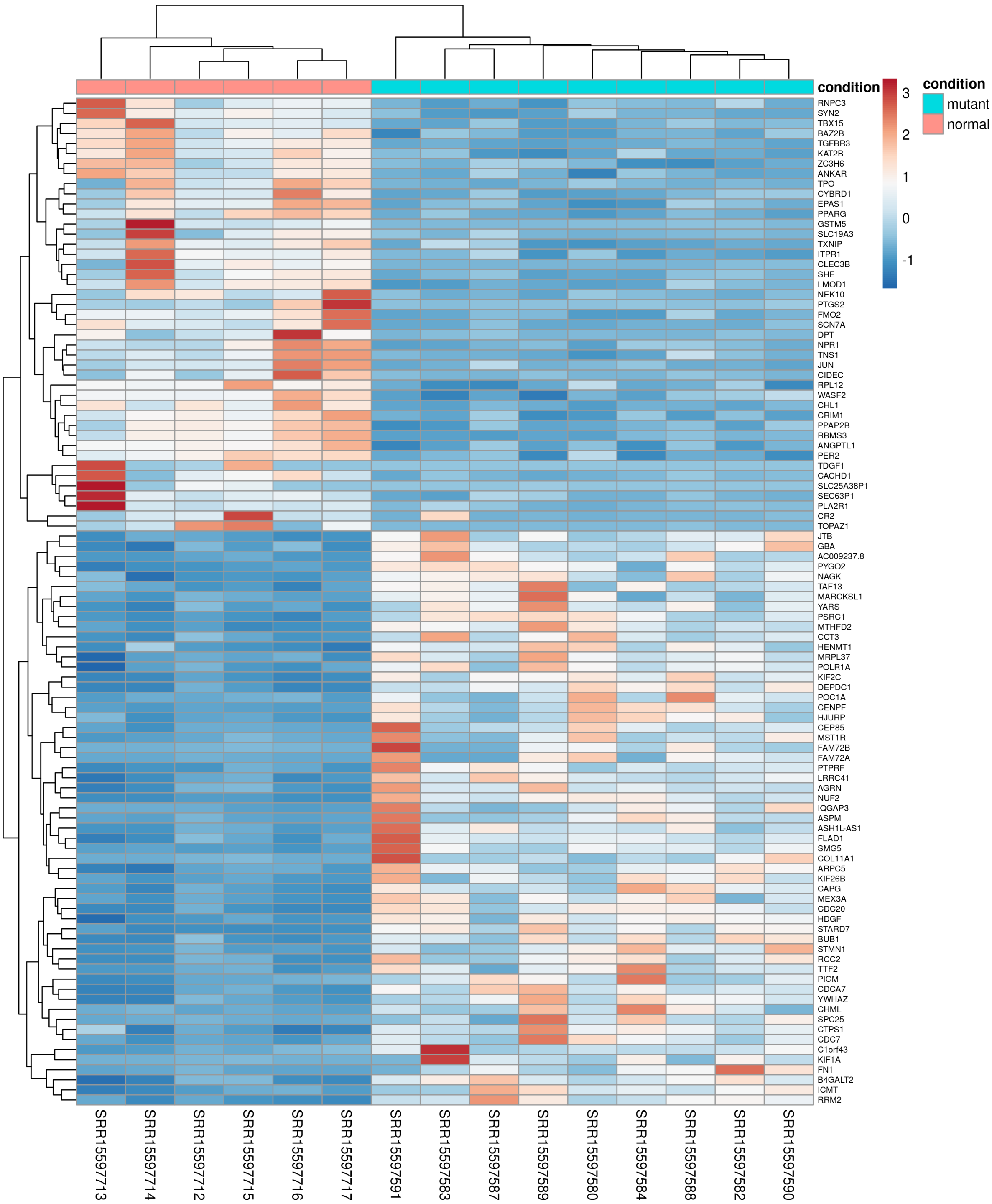


Fig 2: Heatmap based on Top 100 differentially expressed genes between BRCA1 mutant ER -ve tumors vs normal samples

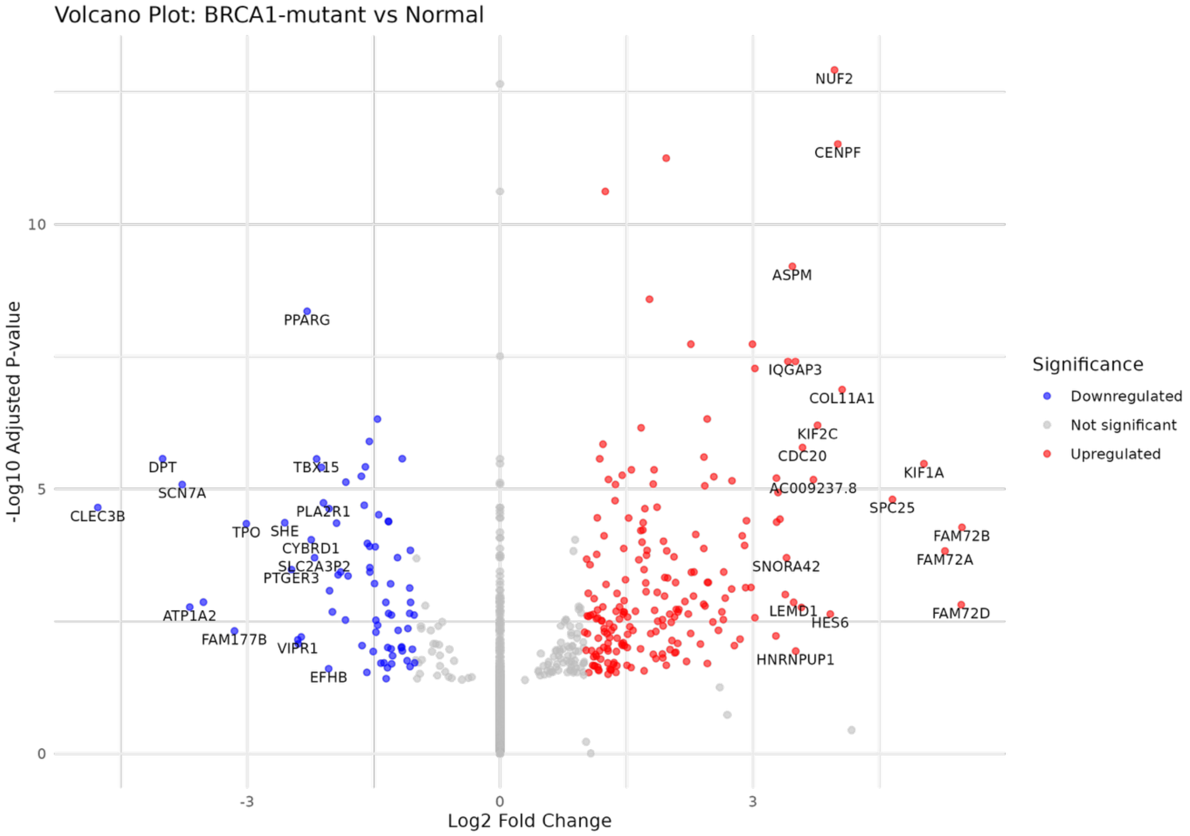


Fig 3: Volcano plot depicting the differentially expressed genes between BRCA1 mutant ER-ve tumors vs normal samples.

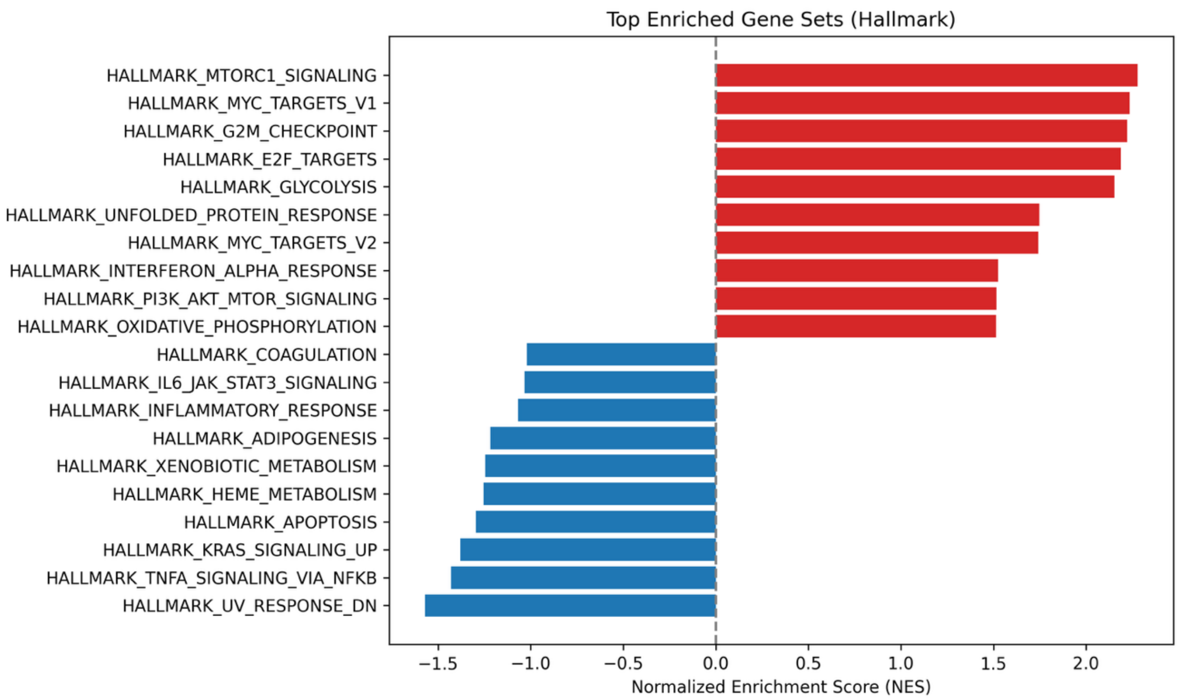


Fig 4: 20 most significantly enriched pathways based on Hallmark genesets (10 upregulated and 10 downregulated)

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